

EXPERIMENT - 24

1. Problem Statement

Write a C program that displays the following menu and uses a switch statement to print the "Increasing pattern" selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

- **Increasing right-angled triangle** — input the number of rows, n , and print an increasing right-angled triangle using asterisks.
- **Increasing row-number triangle** — input n and print a triangle in which each row number is printed as many times as its value.
- **Increasing natural-number pattern** — input n and print natural numbers from 1 to the corresponding row number in each row.
- **Consecutive-number triangle** — input n and print consecutive natural numbers in a triangular pattern.
- **Increasing alphabet pattern** — input n and print uppercase English letters from A up to the letter corresponding to each row. Assume that $1 \leq n \leq 26$.

2. Algorithm

Main Program

Step 1: START

Step 2: Display the menu: 1. Increasing right-angled triangle, 2. Increasing row-number triangle, 3. Increasing natural-number pattern, 4. Consecutive-number triangle, 5. Increasing alphabet pattern.

Step 3: Read choice.

Step 4: Using a switch statement on choice:

 If choice is 1, call `increasingTriangle()`.

 If choice is 2, call `rowNumberTriangle()`.

 If choice is 3, call `naturalNumberPattern()`.

 If choice is 4, call `consecutiveTriangle()`.

 If choice is 5, call `alphabetPattern()`.

Step 5: STOP

Function `increasingTriangle()`

Step 1: Read n .

Step 2: Repeat for $i = 1$ to n :

 Repeat for $j = 1$ to i : print an asterisk.

 Move to a new line.

Step 3: Return to the calling function.

Function `rowNumberTriangle()`

Step 1: Read n .

Step 2: Repeat for $i = 1$ to n :

Repeat for $j = 1$ to i : print the value of i .

Move to a new line.

Step 3: Return to the calling function.

Function `naturalNumberPattern()`

Step 1: Read n .

Step 2: Repeat for $i = 1$ to n :

Repeat for $j = 1$ to i : print the value of j .

Move to a new line.

Step 3: Return to the calling function.

Function `consecutiveTriangle()`

Step 1: Read n .

Step 2: Set $\text{count} = 1$.

Step 3: Repeat for $i = 1$ to n :

Repeat for $j = 1$ to i : print count , then increase count by 1.

Move to a new line.

Step 4: Return to the calling function.

Function `alphabetPattern()`

Step 1: Read n .

Step 2: Repeat for $i = 1$ to n :

Repeat for $j = 1$ to i : print the character obtained by adding $(j - 1)$ to 'A'.

Move to a new line.

Step 3: Return to the calling function.

3. Test Case Table

#	Menu Choice	Input	Expected Output	Actual Output	Result
1	1 — Right-angled triangle	$n = 4$	Rows of 1, 2, 3 and 4 asterisks	As expected	PASS
2	2 — Row-number triangle	$n = 4$	1 / 2 2 / 3 3 3 / 4 4 4 4	As expected	PASS
3	3 — Natural-number	$n = 4$	1 / 1 2 / 1 2 3 / 1 2 3 4	As expected	PASS
4	4 — Consecutive-number	$n = 4$	1 / 2 3 / 4 5 6 / 7 8 9 10	As expected	PASS
5	5 — Alphabet	$n = 4$	A / A B / A B C / A B C D	As expected	PASS
6	1 — Right-angled triangle	$n = 1$	A single asterisk, the smallest triangle	As expected	PASS
7	5 — Alphabet	$n = 26$	26 rows, the last row ending at Z	As expected	PASS

4. Source Code

increasing.c

```
/* Menu-driven program that uses a switch statement to print the
   increasing pattern selected by the user. Each pattern is written in
   its own parameterless function that returns nothing, and every
   pattern is built with nested loops. No arrays are used. */

#include <stdio.h>

void increasingTriangle(void);
void rowNumberTriangle(void);
void naturalNumberPattern(void);
void consecutiveTriangle(void);
void alphabetPattern(void);

int main(void)
{
    int choice;

    printf("1. Increasing right-angled triangle\n");
    printf("2. Increasing row-number triangle\n");
    printf("3. Increasing natural-number pattern\n");
    printf("4. Consecutive-number triangle\n");
    printf("5. Increasing alphabet pattern\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: increasingTriangle(); break;
        case 2: rowNumberTriangle(); break;
        case 3: naturalNumberPattern(); break;
        case 4: consecutiveTriangle(); break;
        case 5: alphabetPattern(); break;
    }

    return 0;
}

/* Pattern 1: row i contains i asterisks. */
void increasingTriangle(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

/* Pattern 2: row i contains the value i printed i times. */
void rowNumberTriangle(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("%d ", i);
        printf("\n");
    }
}

/* Pattern 3: row i contains the natural numbers 1 to i. */
```

```

void naturalNumberPattern(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

/* Pattern 4: the counting continues from one row to the next, so the
   counter is kept outside both loops. */
void consecutiveTriangle(void)
{
    int n, i, j, count;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    count = 1;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) {
            printf("%d ", count);
            count = count + 1;
        }
        printf("\n");
    }
}

/* Pattern 5: row i contains the letters A up to the i-th letter. The
   letter is obtained by adding an offset to the character 'A'. */
void alphabetPattern(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("%c ", 'A' + j - 1);
        printf("\n");
    }
}

```

5. Compilation and Output

The program was compiled with `gcc -Wall -Wextra increasing.c -o increasing` and it compiled without a single warning. The screenshots below are of the actual runs on my machine.

Patterns 1 and 2

```
adilshakil — zsh — 127x32
adilshakil@ADILs-MacBook-Air ~ % cd ~/Desktop/MyAssignment
adilshakil@ADILs-MacBook-Air MyAssignment % gcc -Wall -Wextra increasing.c -o increasing
adilshakil@ADILs-MacBook-Air MyAssignment % ./increasing
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter the number of rows: 4
*
* *
* * *
* * * *
adilshakil@ADILs-MacBook-Air MyAssignment % ./increasing
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter the number of rows: 4
1
2 2
3 3 3
4 4 4 4
█
```

Patterns 3 and 4

```
adilshakil — zsh — 127x32
adilshakil@ADILs-MacBook-Air MyAssignment % ./increasing
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter the number of rows: 4
1
1 2
1 2 3
1 2 3 4
adilshakil@ADILs-MacBook-Air MyAssignment % ./increasing
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter the number of rows: 4
1
2 3
4 5 6
7 8 9 10
█
```

Pattern 5, and the Smallest Value of n

```
adilshakil — zsh — 127x32
adilshakil@ADILs-MacBook-Air MyAssignment % ./increasing
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter the number of rows: 4
A
A B
A B C
A B C D
adilshakil@ADILs-MacBook-Air MyAssignment % ./increasing
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter the number of rows: 1
*
```

Alphabet Pattern at n = 26

```
adilshakil — zsh — 127x32
Enter your choice: 5
Enter the number of rows: 26
A
A B
A B C
A B C D
A B C D E
A B C D E F
A B C D E F G
A B C D E F G H
A B C D E F G H I
A B C D E F G H I J
A B C D E F G H I J K
A B C D E F G H I J K L
A B C D E F G H I J K L M
A B C D E F G H I J K L M N
A B C D E F G H I J K L M N O
A B C D E F G H I J K L M N O P
A B C D E F G H I J K L M N O P Q
A B C D E F G H I J K L M N O P Q R
A B C D E F G H I J K L M N O P Q R S
A B C D E F G H I J K L M N O P Q R S T
A B C D E F G H I J K L M N O P Q R S T U
A B C D E F G H I J K L M N O P Q R S T U V
A B C D E F G H I J K L M N O P Q R S T U V W
A B C D E F G H I J K L M N O P Q R S T U V W X
A B C D E F G H I J K L M N O P Q R S T U V W X Y
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
```

All seven test cases gave the expected result, and the five patterns match the sample outputs given in the question exactly.